



DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

(An Institution of National Importance under the Ministry of Education, Govt of India)

MID SEMESTER EXAMINATIONS-I

M&C IV Semester

MAPC-206: Theory of Computation

Time: 04:00 PM to 04:50 PM

Max. Marks: 15

Date: 20-02-2026

N.B.: Answer ALL questions. Each question carries equal marks.

Answers to all parts of each question should be in one place.

Q1	Write a language of strings containing substring 'ab' over the alphabet $\Sigma = \{a, b\}$. Write its regular expression. Is regular expression of the above language unique? Write its NFA and DFA. Is NFA for the above language unique? Write the grammar of the above language. Is grammar of the above language unique? Can NFA and DFA for the above language be same?																		
Q2	Write NFA and DFA for the language $L = \{ab, aba\}$ over the alphabet $\Sigma = \{a, b\}$. Find its regular expression. Find the minimum number of states in NFA and DFA. Find the minimum number of states in its finite automata.																		
Q3	Minimize the following DFA whose transition table is given below. <table><tr><th>State</th><th>Input 0</th><th>Input 1</th></tr><tr><td>A</td><td>B</td><td>C</td></tr><tr><td>B</td><td>B</td><td>D</td></tr><tr><td>C</td><td>B</td><td>C</td></tr><tr><td>D</td><td>B</td><td>E</td></tr><tr><td>E</td><td>B</td><td>C</td></tr></table>	State	Input 0	Input 1	A	B	C	B	B	D	C	B	C	D	B	E	E	B	C
State	Input 0	Input 1																	
A	B	C																	
B	B	D																	
C	B	C																	
D	B	E																	
E	B	C																	